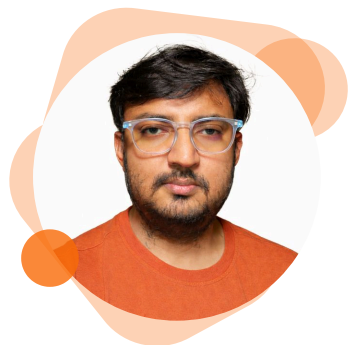




SOURAV MONDAL

AI/ML SCIENTIST | COMPUTATIONAL
CHEMIST | DRUG DISCOVERY EXPERT

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- <https://mondalsou.github.io/mondalsou/>
- Hyderabad, India

FIND ME ONLINE

Portfolio

<https://mondalsou.github.io/mondalsou/>

Github

<https://github.com/mondalsou>

LinkedIn

<https://www.linkedin.com/in/sourav1/>

SUMMARY

AI/ML Scientist | 10+ years in drug discovery, computational chemistry, and deep learning. Core contributor to Igniva™ Agentic AI platform at Aganitha, building production GNN pipelines, AI Agents, and QM tools for CSP, conformer generation, pKa, and ADMET. PhD (JNCASR) + Postdoc (Trinity College Dublin). Published in Nature Comput. Mat., JACS, Nano Letters. PyTorch, GNNs, RDKit, DFT, Azure.

KEY ACHIEVEMENTS

6+ Top-Tier Publications

Published in Nature Computational Materials, JACS, ACS Cent. Sci., and Nano Letters as first or equal-contribution author.

EXPERIENCE

Aganitha's Agentic AI platform for biopharma R&D

Hyderabad, India

Aganitha Cognitive Solutions Scientist

01/2024 - Present

- Core contributor to Igniva™ - Aganitha's Agentic AI platform for biopharma R&D, developing multiple production tools for the small molecule design and CMC pipeline
- Developed Crystal CleanPro and CrystalLatticeAI - Crystal Structure Prediction (CSP) pipelines leveraging Graph Neural Networks for rapid polymorph screening of drug-like molecules
- Built EAGLE - AI-driven platform for 3D molecular conformer generation, enabling environment-aware drug discovery applications
- Engineered SitepKa - site-specific pKa prediction tool using dual-graph cross-attention GNN architecture for non-aqueous conditions
- Developed QM-based pKa estimation pipeline using first-principles quantum chemistry methods, complementing the GNN approach for high-accuracy predictions
- Created physics-based QM platform for ADMET property prediction including solubility, logP, and solvation energy with production-grade deployment on Azure
- Managed client communications, project proposals, and end-to-end delivery of computational chemistry solutions for pharmaceutical partners

QpiVolta Technologies

Bangalore, India

Computational Scientist & Deep Learning Engineer

08/2023 - 12/2023

- Developed machine learning force fields (MLFF) for solid-state electrolyte materials, enabling rapid molecular dynamics simulations
- Modeled thermodynamic properties of electrolyte systems using molecular dynamics and ML-augmented simulations
- Led contract research on complex biological system simulations, delivering actionable insights to stakeholders

Trinity College Dublin

Dublin, Ireland

Postdoctoral Researcher

03/2021 - 07/2023

- Built neural network models for quantum property prediction of paramagnetic defects, published in Nature Computational Materials (npj Comput. Mat., 2023)
- Developed first-principles computational methods for spin-phonon dynamics simulation, published in JACS (2022)
- Combined ab-initio quantum chemistry methods with machine learning to model spin-lattice relaxation in single-molecule magnets
- Collaborated with experimental groups across Europe on quantum computing and molecular magnetism research



KEY ACHIEVEMENTS

● 6+ Production Pipelines in Igniva™

Built CrystalCleanPro, CrystalLatticeAI, EAGLE, SitepKa, QM-pKa, and ADMET platforms — shipped as part of Aganitha's commercial Agentic AI product.

● AI Agent for Lead Optimization

Developed autonomous medicinal chemistry agent using Claude AI with iterative propose-score-compare loops for drug lead optimization.

● End-to-End Drug Discovery Platform

Built and deployed full-stack triage app(React, FastAPI, RDKit) with real-time ADMET, QED, and PAINS screening on Vercel.



SKILLS

AI/ML & Deep Learning

AI/ML & Deep Learning ·
Graph Neural Networks (GNN) ·
PyTorch · TensorFlow · Transformers ·
AI Agents · Neural Networks ·
Transfer Learning ·
Hyperparameter Tuning · MLOps ·
CI/CD · Reinforcement Learning ·
Multi-tasking Learning · LSTM ·
LangChain · Bayesian Optimization ·
RAG

Drug Discovery & Cheminformatics

ADMET Prediction ·
Molecular Property Prediction ·
Conformer Generation ·
pKa Prediction · CADD · RDKit ·
SMILES/SMARTS · Molecular Docking ·
QED Scoring · Lead Optimization ·
Binding Affinity Prediction ·
Formulation DoE



PERSONAL PROJECTS

SitepKa – Site-Specific pK_a Prediction

Dual-Graph Cross-Attention GNN · PyTorch · RDKit · 35 Industrial Solvents

- Site-specific pKa prediction across water and 34 industrial organic solvents using a dual-graph cross-attention GNN that explicitly models solute-solvent interactions, supplemented by Kamlet-Taft solvent descriptors
- Trained on 33,269 experimental measurements from 10 public sources
- Achieves competitive aqueous accuracy (MAE 1.49–2.30, SAMPL6–8 blind challenges) and transfers effectively to non-aqueous solvents including DMF and pyridine (MAE 0.87–0.99) — a regime where deep-learning tools are largely absent

<https://chemrxiv.org/doi/full/10.26434/chemrxiv.15002800/v2>

Lead Optimization Agent

AI Agents · LangChain · RDKit · Streamlit · Hermes

- Built AI-powered medicinal chemistry sandbox using LangChain for autonomous iterative lead optimization with propose-score-compare agent loop
- Tracks BBB permeability, CNS MPO, QED, and molecular flexibility per iteration with highlighted 2D structural changes
- Deployed on HuggingFace Spaces with local run persistence for cost-efficient multi-session workflows

https://huggingface.co/spaces/mondalsou/lead_optimization_agent

GNNBindOptimizer

PyTorch · HGTCnv · REINFORCE · SQL Server · MLflow · Docker

- Heterogeneous GNN (HGTCnv) for protein-ligand binding affinity prediction using multi-task learning with Kendall uncertainty-weighted loss across affinity, pose classification, and selectivity objectives.
- Reinforce-based molecular generator targeting EGFR kinase pocket, with generated candidates scored and benchmarked against AutoDock Vina docking.
- End-to-end deployment via Docker Compose with SQL Server persistence, MLflow experiment tracking, and Streamlit UI covering binding prediction, RL generation, and live GNN-vs-Vina comparison.

<https://github.com/mondalsou/gnn-bind-optimizer>

Formulation Bayesian Optimization

BoTorch · GPyTorch · Gaussian Process · Streamlit

- Model-based DoE pipeline replacing OFAT/grid screening with GP surrogate models for pharmaceutical tablet formulation, optimizing dissolution under hard mass-balance constraints across a 5-exipient design space
- Implements single-objective (GP + EI/LogEI) and multi-objective (qNEHVI Pareto front) Bayesian optimization over dissolution, hardness, and friability with uncertainty-aware sequential experimentation

<https://github.com/mondalsou/formulation-bayesian-optimization>

RegDocIQ – Regulatory Document Q&A

BM25 · pypdf · OpenRouter LLM · Streamlit

- Minimal RAG pipeline for pharmaceutical regulatory PDF Q&A using BM25 keyword retrieval over overlapping page chunks — zero embedding cost, interpretable ranking as an alternative to vector-DB RAG
- LLM-generated answers with cited source passages via OpenRouter; local-first Streamlit deployment with no cloud dependency

<https://github.com/mondalsou/regdociq>

SKILLS

Computational Chemistry

DFT • Quantum Chemistry • CP2K •
ORCA • Psi4 • PySCF • SAPT •
ML Force Fields • MD •
First-Principles Calculations

Programming & Tools

Python • Fortran • NumPy • Pandas •
Scikit-learn • ASE • Docker •
Microsoft Azure • Git • MLflow • n8n •
Streamlit • SQL Server • Google ADK •
OpenClaw • Hermes

EDUCATION

Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)

Bangalore, India

PhD in Computational Materials Science 01/2015 - 02/2021

- Thesis: Tailoring Properties of 2D Systems via Molecular Adsorption and Defect Engineering

Indian Institute of Technology (IIT) Guwahati

Guwahati, India

M.Sc in Chemistry 01/2012 - 12/2014

- Qualified NET CSIR-UGC Junior/Senior Research Fellowship

PUBLICATIONS

SitepKa: Site-Specific Acid and Base pKa Prediction Across Multiple Industrial Solvents

2026

ChemRxiv

S. Mondal, A. Halder and C. Nanduru

The spin phonon relaxation of single molecules magnet in the presence of strong exchange coupling

2025

ACS Cent. Sci.

S. Mondal, J. Netz et al.

Spin-phonon decoherence in solid-state paramagnetic defects from first principles

2023

Nature Computational Materials

S. Mondal and A. Lunghi

Unravelling the contributions to spin-lattice relaxation in Kramers single-molecule magnets.

2022

JACS

S. Mondal and A. Lunghi